

# Charles O'Byrne

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Eagle Scout

## EDUCATION

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### University of Iowa

*Iowa City, IA*

*Graduation: May 2027*

*Bachelor of Science in Engineering*

*Expected Minors: Math, Computer Science,*

*Electrical Engineering: Computer*

*Artificial Intelligence*

## ENGINEERING EXPERIENCE

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### Summer Internship

*Itasca, IL*

*June 2025 – August 2025*

*Leister Technologies*

- Conducted research and executed test runs to evaluate and implement automation solutions using Excel VBA, Javabased macros, and GUI tools in HTML/JavaScript, whilst paired with AI to improve internal software processes.
- Collaborated with engineering teams through tool repair trainings and technical sessions, gaining hands-on exposure to product design, diagnostics, and industrial heating tool applications.
- Represented the company at major trade show SUR/FIN, networking with partner firms and industry leaders to expand professional connections and strengthen technical knowledge of emerging engineering technologies.

### Computer Builder & Consultant

*Oakbrook, IL*

*May 2020 – Present*

*Personal Business*

- Assembled and configured 30+ custom-built computers, integrating ideal hardware components to meet diverse performance and budget needs while ensuring exceptional customer satisfaction.
- Conducted 10+ educational workshops, guiding individuals of varying skill levels in building computers from scratch.
- Spearheaded multiple client projects simultaneously which focused on overclocking and optimizing computer hardware, resulting in improved system performance and reliability.

### Hand-Voice Controlled Drone System

*Iowa City, IA*

*April 2026 – May 2026*

*IoT Drone Control & Simulation Project*

- Built an IoT-based drone simulation system using Arduino Nano 33 BLE, Raspberry Pi, Webots, and the Crazyflie quadcopter model.
- Developed a gesture-control pipeline that processed IMU roll, pitch, and yaw data with Madgwick filtering and transmitted motion commands over BLE and UDP.
- Integrated microphone-based whistle input for altitude control and logged drone telemetry, including position and orientation data, to InfluxDB for flight-path analysis.

## WORK EXPERIENCE

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### Pool Manager

*Oak Brook, IL*

*August 2019 – Present*

*Oak Brook Park District Aquatic Center*

- Manage a team of lifeguards and oversee all pool operations, including water chemistry, pump room oversight, and facility maintenance, while contributing to management meetings for strategic facility planning.
- Ensure a safe and welcoming environment by addressing patron needs, resolving issues, and upholding high customer service standards, prioritizing safety and customer satisfaction.

### IT Contractor

*Tinley Park, IL*

*July 2021 – August 2026*

*EPS Rehabilitation Inc*

- Disassembled 7 malfunctioning computers and reconfigured them into 4 operational systems, optimizing resource utilization.
- Performed maintenance on existing computers, updating software and installing new applications to enhance system performance and reliability.

## SKILLS

Embedded

Breadboarding

C/C++

Assembly

Java

Python

HDL/FPGA